

EXPLORING GENERATIVE AI FOR STUNNING IMAGES



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Experience **3 Cutting-Edge AI Tools** using which, you can:

- Turn your imagination to reality with just a few lines of text
- Create stunning photorealistic images
- Generate personalized avatars
- Edit images like a pro!

Date: 19th October
Time: 10 - 11 AM (IST)

A little about me...

- **Microsoft | Bing Autosuggest** (2021 – Present)
 - Data & Applied Scientist II (2023 - Present)
 - Data & Applied Scientist (2021 - 2023)
- **IIT Bombay** (2017 – 21)
 - Major in Mechanical Engineering (9.90 CPI)
 - Minor in Computer Science & Engineering
 - Department Rank 1 | Institute Rank 2
- **AIR 1 – ICSE** (2015)
- Amazon **#1 Best Selling Author** of “Beyond Code”
- Holder of **1 US Patent**
- **Areas of Interest:** AI Tools, NLP, Data Analysis



Before we begin, here's a question...

Which is real & which is AI?



A



B

Both are AI!

Generated by me last night 😊

That's where we are today!

AI is truly blurring the boundaries between imagination & reality,
empowering us to explore limitless possibilities!

Agenda

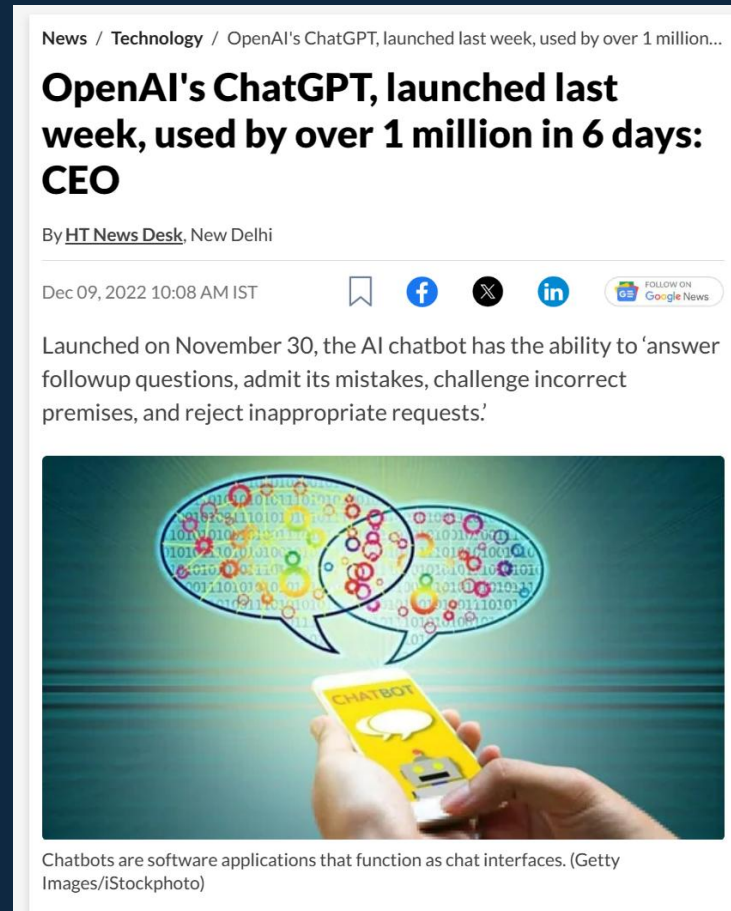
- The Generative AI Revolution
- Prompt Engineering 101
- Stunning Images with Dall-E 3
 - Turning Imagination into Reality
 - Generating Photorealistic Images
- Editing like a Pro with ClipDrop
 - Uncrop, Cleanup, BG Removal, etc.
 - Sketch-To-Image using Stable Doodle
- AI Art – A Myriad of Opportunities
- Quiz Time!



The Generative AI Revolution



April 2022



December 2022



February 2023

... since then, AI has been dominating the industry, changing every paradigm of work with new Models, Tools & Startups almost every week!



WE GAVE THEM EVERYTHING

**This movie trailer was generated by
Norman Neubert, completely using AI!**



Welcome to the Future!

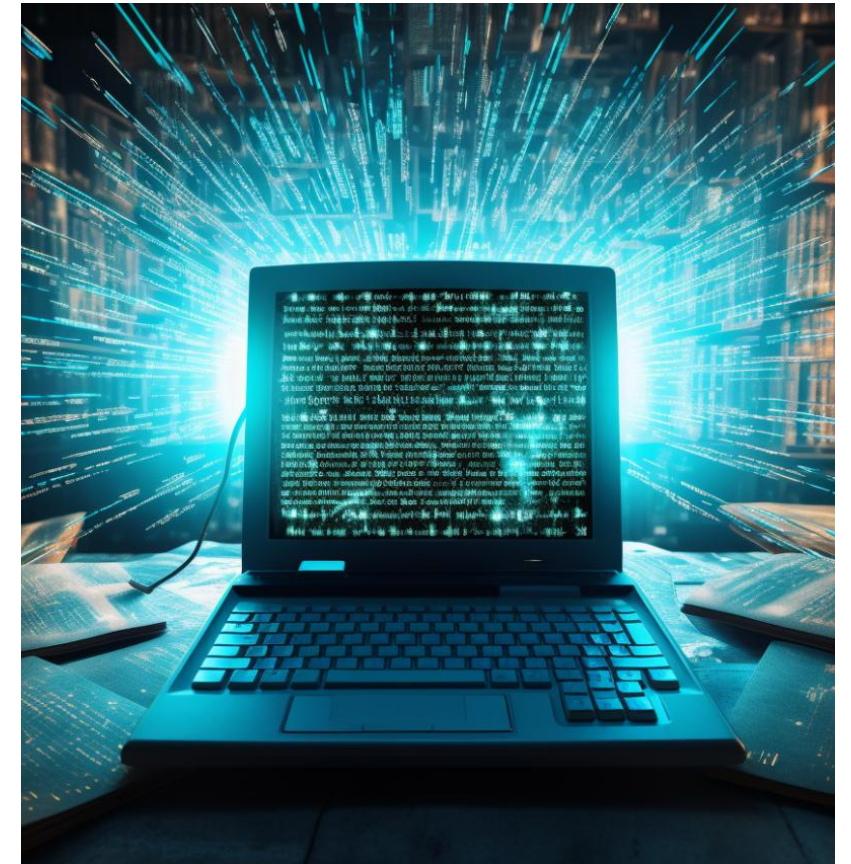
Let's take our first step towards creating something really cool with AI...



Prompt Engineering 101

What is Prompt Engineering?

- A ***prompt*** is a set of instructions that tells the AI what you wish to accomplish
- Prompt engineering is the ***‘art of communicating with AI’***
- AI will try to understand your prompt and create an image that matches it



Why 'Prompt Engineering'?



- Helps us to *explore our creativity and imagination*, by giving us the power to create any kind of image we can think of
- Helps us to get *better results from the AI model*, by making our expectations clear and specific
- Highly **in-demand skill**, useful wherever Gen-AI finds applications

Jumpstart your creativity

The right words can be hard to find. Here are a few ideas to get you started.

Vibes / Mood
Try including a time of day, a feeling, or aesthetic.

+ golden hour + playful

Photography
Try including a time of day, a feeling, or aesthetic.

+ 85mm lens + wide angle

Art Styles
Try an illustration style, art period or material.

+ marble + impressionism

You have 100 boosts for today!

Use your boosts to create images more quickly. If you run out, image generation may take longer.

×

You can now create images in the new Bing.

Learn more

Explore ideas

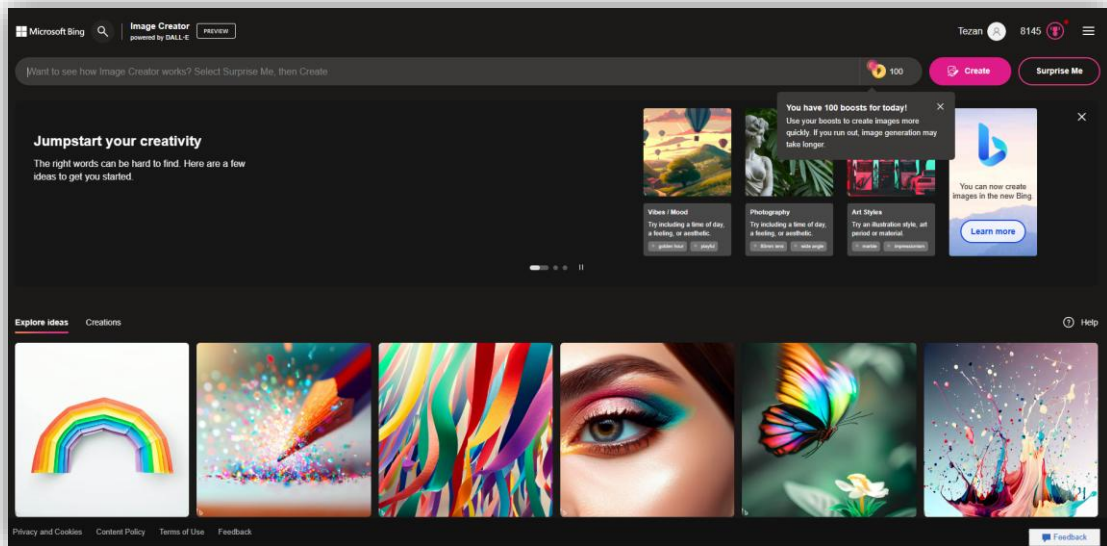
Creations

Stunning Images with Dall-E 3

Bring Your Imagination to Life...

- Text2Image AI Model
- Accessed using Bing Image Creator
- 100 boosts on signup

- Perfect for:
 - Projects
 - Presentations (*all images in this PPT* 😊)
 - Book Covers
 - Comics
 - ...and much more



<https://www.bing.com/create>



(Realistic) streets of Southern France, summertime, bright and beautiful, cinematic landscape



(Crazy Concepts) An astronaut surfing on the moon with Earth in the background

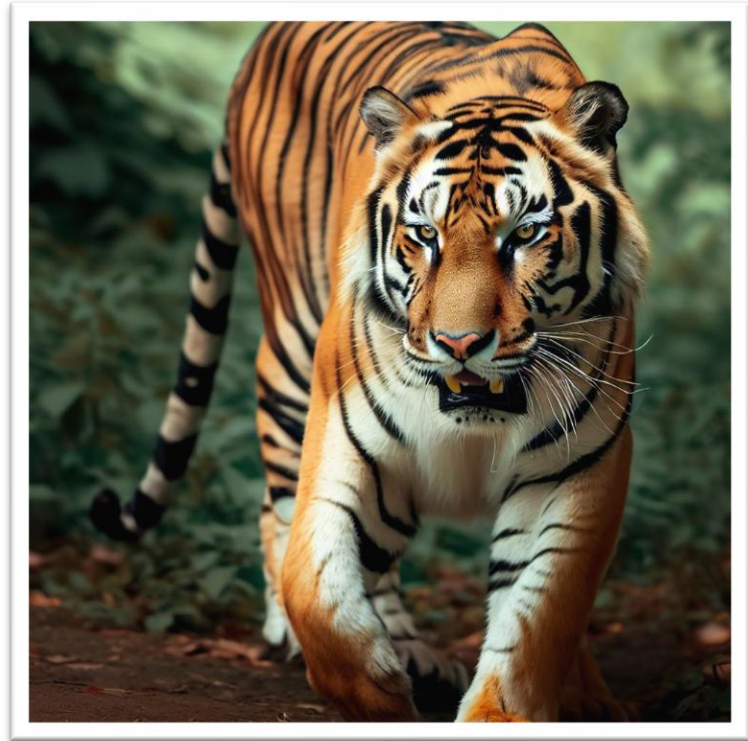
Your images can be realistic, or even some vague concept...

Concept Art



a cute robot scuba diving near a coral reef, colorful & vibrant underwater creatures & plants, digital art

Photorealistic Images



Royal Bengal tiger walking in a forest

Let's get into some action, with Concept Artworks!

Book Cover



The Lion, the Witch and the Wardrobe, fantasy book cover background, intricate details, 8k, digital art

Comics Book Artwork



Two samurais running towards each other in an epic battle clash, in the style of comic book drawing

Funky Cards for Special Occasions



kid cutting birthday cake, surrounded by avengers clapping, celebratory background, balloons with numbers 10 behind

Generating Realistic Images

Prompt Template

[Subject] in the ***[style/technique]***, set in a ***[environment/setting]*** with a ***[mood/atmosphere]***. Include ***[details/modifiers]***, emulate a ***[specific camera]*** and ***[aperture]***, use ***[lighting conditions]***, and compose with ***[composition/perspective guidelines]***.

But how do I get all this
technical information & jargon?

Well...leave that to Bing Chat!

Simple vs Detailed Prompt Results

Simple Prompt



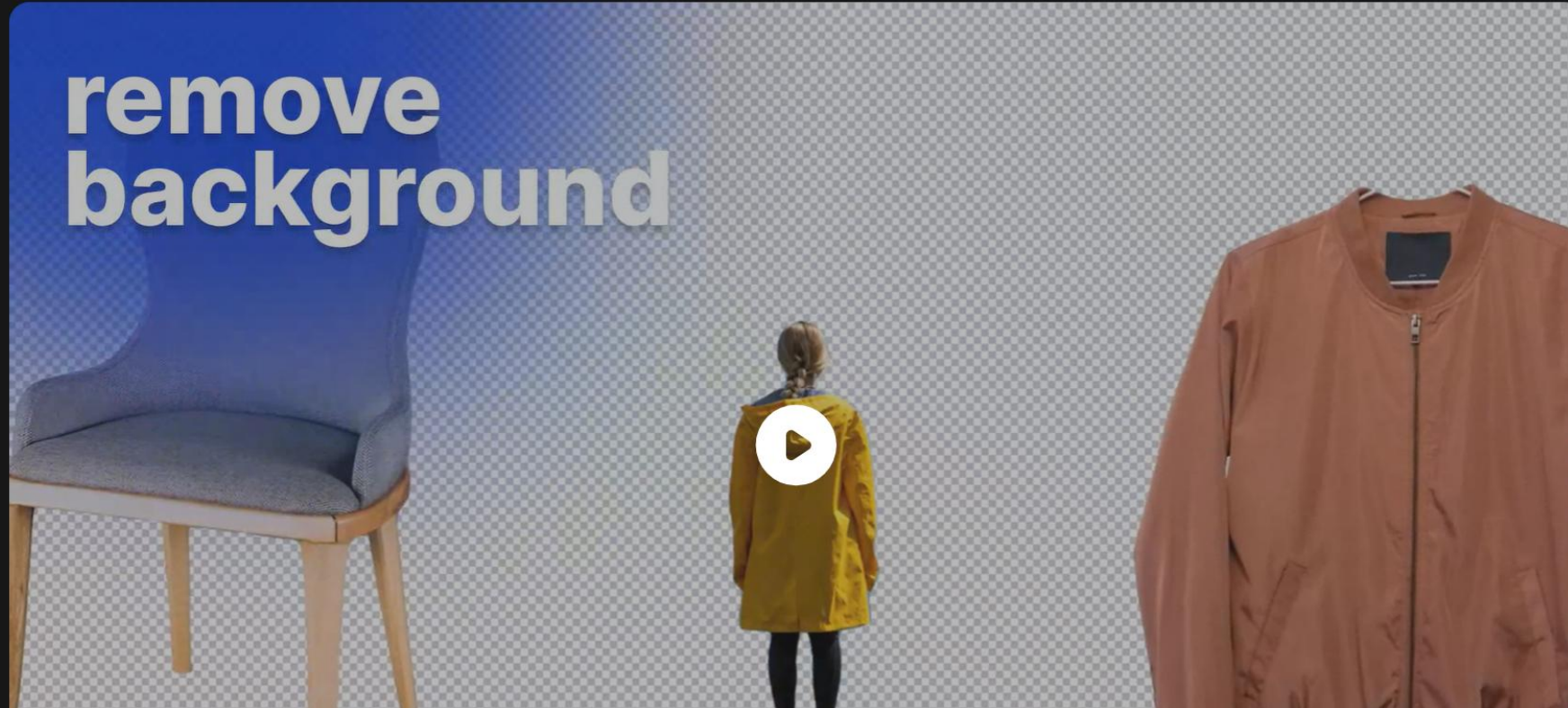
Royal Bengal tiger walking in a forest

Detailed Prompt



Royal Bengal tiger walking in a forest, painted in photorealistic style with oil on canvas, set in a rainy autumn environment with a somber mood. Include details such as the tiger's stripes and the raindrops. Emulate a Canon EOS 5D Mark IV camera with an aperture of f/2.8, use natural lighting conditions, and compose with the rule of thirds.

CREATE STUNNING VISUALS IN SECONDS



Editing Like a Pro with ClipDrop

Suite of AI Image Tools in ClipDrop (<https://clipdrop.co/>)



Uncrop

Uncrop your photos to any image format



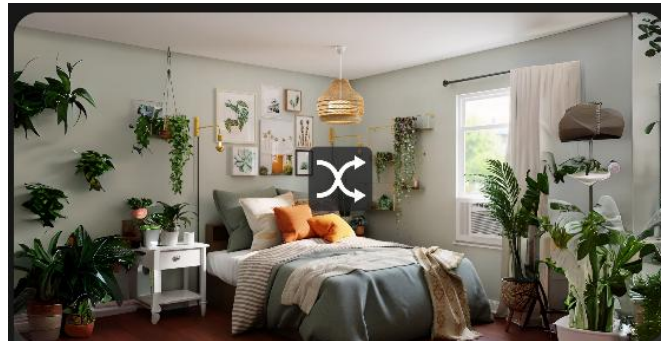
Cleanup

Remove objects, people, text and defects from your pictures automatically.



Text remover

Remove text from any image



Reimagine XL

Create multiple variants of an image with Stable Diffusion



Relight

Relight your images with beautiful lights.

Sketch-To-Image with Stable Doodle

Sketch



Variety of
Prompts



Image(s)



Let's summarize what we've seen till this point...

- The rise of Generative AI Revolution
- The “what’s” & “why’s” of Prompt Engineering
- Creating absolutely stunning images with Dall-E 3
 - Concept Art
 - Photorealistic Images
- Exploring StableDiffusion in ClipDrop
 - Buffet of Image Editing Tools
 - Sketch2Image using Stable Doodle



AI Art – A Myriad of Opportunities

Art with Generative AI can be applied in several compelling use-cases...

- Creating digital art
- Designing book covers
- Creating comics
- Designing video game characters
- Creating personalized cards
- Designing logos
- Creating custom t-shirts
- Designing posters
- Creating animations
- Designing album covers
- Creating memes
- Creating digital collages
- Designing social media posts
- Creating digital scrapbooks
- Designing infographics
- Creating personalized phone cases
- Designing custom calendars
- Creating digital storybooks
- Designing custom mugs
- **Designing stickers!**

It's Quiz Time!!!



GO TO
menti.com

ENTER THE CODE
2911 9183

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That's All Folks!

For More Interesting Stuff on AI , Data Science & the Latest Technologies...



TEZAN SAHU

Reach Out to me for
Speaker Sessions